

Curve fitting algorithm for multimodal particle size distributions – a theoretical basis

2025-11-20

Contents

1 Setup	1
1.1 Retrieve Read Functions	1
1.2 Import Libraries	1
1.3 User Paths	2
2. Formatting Data	2
2.1 Brechtel Manufacturing Inc. (BMI) Data	2
2.2 TSI Data	2
2.3 netCDF Data	2
2.4 NASA-AMES Data	3
3. Running multimodal	3
3.1 Example 1 - Laboratory Data	3
3.2 Example 2 - Storm Peak Laboratory	7

Christopher N. Rapp¹, Sining Niu², Yue Zhang², A. Gannet Hallar³, Fred J. Brechtel⁴, and Daniel J. Cziczo¹

¹Department of Earth, Atmospheric, and Planetary Sciences, Purdue University, West Lafayette, Indiana, 47906-2051, USA

²Department of Atmospheric Sciences, Texas A&M University, College Station, Texas, 77843-3150, USA

³Department of Atmospheric Sciences, University of Utah, Salt Lake City, Utah, 84112-0102, USA

⁴Brechtel Manufacturing Incorporated, Hayward, California, 94544, USA

Correspondence to: Christopher N. Rapp (christopherrapp@icloud.com)

Source code for helper read functions are housed here <https://github.com/christopher-rapp/multimodal.git>

1 Setup

1.1 Retrieve Read Functions

1.2 Import Libraries

```
library(stringr)
library(data.table)
library(tidyr)
```

```
library(dplyr)
library(lubridate)
library(logr)
library(ggplot2)
library(purrr)
library(patchwork)
library(ncdf4)
```

1.3 User Paths

To download the example data for BMI and TSI visit. For NASA-AMES or netCDF examples see linked sources below.

<https://github.com/christopher-rapp/multimodal/tree/main/data>

Define user paths such as '~/Library/Documents/multimodal/example/BMI/'

2. Formatting Data

Raw view of data format is appended to the bottom of this document

2.1 Brechtel Manufacturing Inc. (BMI) Data

```
BMI.data.ls <- readPSD_BMI(import.path.BMI, tz = "US/Eastern")

# Read functions export data as a list to account for multiple files in a directory
dataPSD.BMI <- BMI.data.ls[[1]]
```

2.2 TSI Data

Use example data from Storm Peak Laboratory for a new particle formation event (NPF) on 2022-03-23.

```
TSI.data.ls <- readPSD_TSI(import.path.TSI, tz = "US/Mountain")

# Read functions export data as a list to account for multiple files in a directory
dataPSD.TSI <- TSI.data.ls[[1]]
```

2.3 netCDF Data

Data were obtained from the Atmospheric Radiation Measurement (ARM) User Facility, a U.S. Department of Energy (DOE) Office of Science user facility managed by the Biological and Environmental Research Program.

SGP SMPS data were obtained from the Atmospheric Radiation Measurement (ARM) User Facility, a U.S. Department of Energy (DOE) Office of Science user facility managed by the Biological and Environmental Research Program.

Kuang, C., Singh, A., Howie, J., Salwen, C., & Hayes, C. Scanning mobility particle sizer (AOSSMPS), 2016-11-15 to 2025-06-23, Southern Great Plains (SGP), Lamont, OK (Extended and Co-located with C1) (E13). Atmospheric Radiation Measurement (ARM) User Facility. <https://doi.org/10.5439/1476898>

Similarly for the BMI and TSI data, define the import.path of downloaded NC

```
NC.data.ls <- readPSD_NC(import.path.NC)
```

```
# Read functions export data as a list to account for multiple files in a directory
dataPSD.NC <- NC.data.ls[[1]]
```

2.4 NASA-AMES Data

Data obtained from <https://ebas-data.nilu.no/DataSets.aspx?stations=US9050R&InstrumentTypes=smps&fromDate=1970-01-01&toDate=2025-12-31>.

The EBAS database has largely been funded by the UN-ECE CLRTAP (EMEP), AMAP and through NILU internal resources. Specific developments have been possible due to projects like EUSAAR (EU-FP5)(EBAS web interface), EBAS-Online (Norwegian Research Council INFRA) (upgrading of database platform) and HTAP (European Commission DG-ENV)(import and export routines to build a secondary repository in support of www.htap.org). A large number of specific projects have supported development of data and meta data reporting schemes in dialog with data providers (EU)(CREATE, ACTRIS and others).

```
NAS.data.ls <- readPSD_NAS(import.path.NAS)
```

```
# Read functions export data as a list to account for multiple files in a directory
dataPSD.NAS <- NAS.data.ls[[1]]
```

3. Running multimodal

Let's run multimodal on an example dataset using a Brechtel SEMS (Model 2002). Note the log path will need to be changed to whatever location you'd like it sent to!

3.1 Example 1 - Laboratory Data

```
# Frequency is null here because I already grouped data above
# I've also removed the argument for the log.path which will default
# to a temporary directory created by R which is deleted upon exiting an R session
result <- multimodal.fitting(dataPSD.BMI,
```

```
    frequency = NULL,
    labeling = T,
    max.iterations = 30,
    max.modes = 6,
    smoothing = F,
    lower.limit = 10,
    upper.limit = 1500,
    NMRSE.threshold = 0.05,
    FVU.threshold = 5,
    FVU.tolerance = 0.1,
    verbose = T)
```

```
## [1] "Log Path: /var/folders/2j/rmf9p1l50wg904hz8qkpv4m0000gn/T//Rtmpv6RxSf/multimodal20231031181523"
## [1] "Current Dataset Time: 2023-10-31 22:15:23 UTC"
## [1] "Dataset sampling frequency is 2.4 min"
## [1] "2023-10-31 22:15:23: Current Loop Iteration: 1, Remaining Variance: 94.93%, Number of Modes: 1"
## [1] "2023-10-31 22:15:23: Current Loop Iteration: 2, Remaining Variance: 3.93%, Number of Modes: 2"
## [1] "2023-10-31 22:15:23: Current Loop Iteration: 3, Remaining Variance: 2.57%, Number of Modes: 3"
## [1] "2023-10-31 22:15:23: Current Loop Iteration: 4, Remaining Variance: 1.63%, Number of Modes: 4"
## [1] "2023-10-31 22:15:23: Current Loop Iteration: 5, Remaining Variance: 1.34%, Number of Modes: 5"
## [1] "2023-10-31 22:15:23: Current Loop Iteration: 6, Remaining Variance: 1.3%, Number of Modes: 6"
## [1] "Concentration RMSE: 2132.57 n/cc"
```

```
# As there is only 1 data file for this set, we will flatten the list  
result <- flatten(result)
```

3.1.1 Outputs

The first element of result is a pass flag i.e. T or F

```
result$pass
```

```
## [1] TRUE
```

The second is the plot which consists of three panels

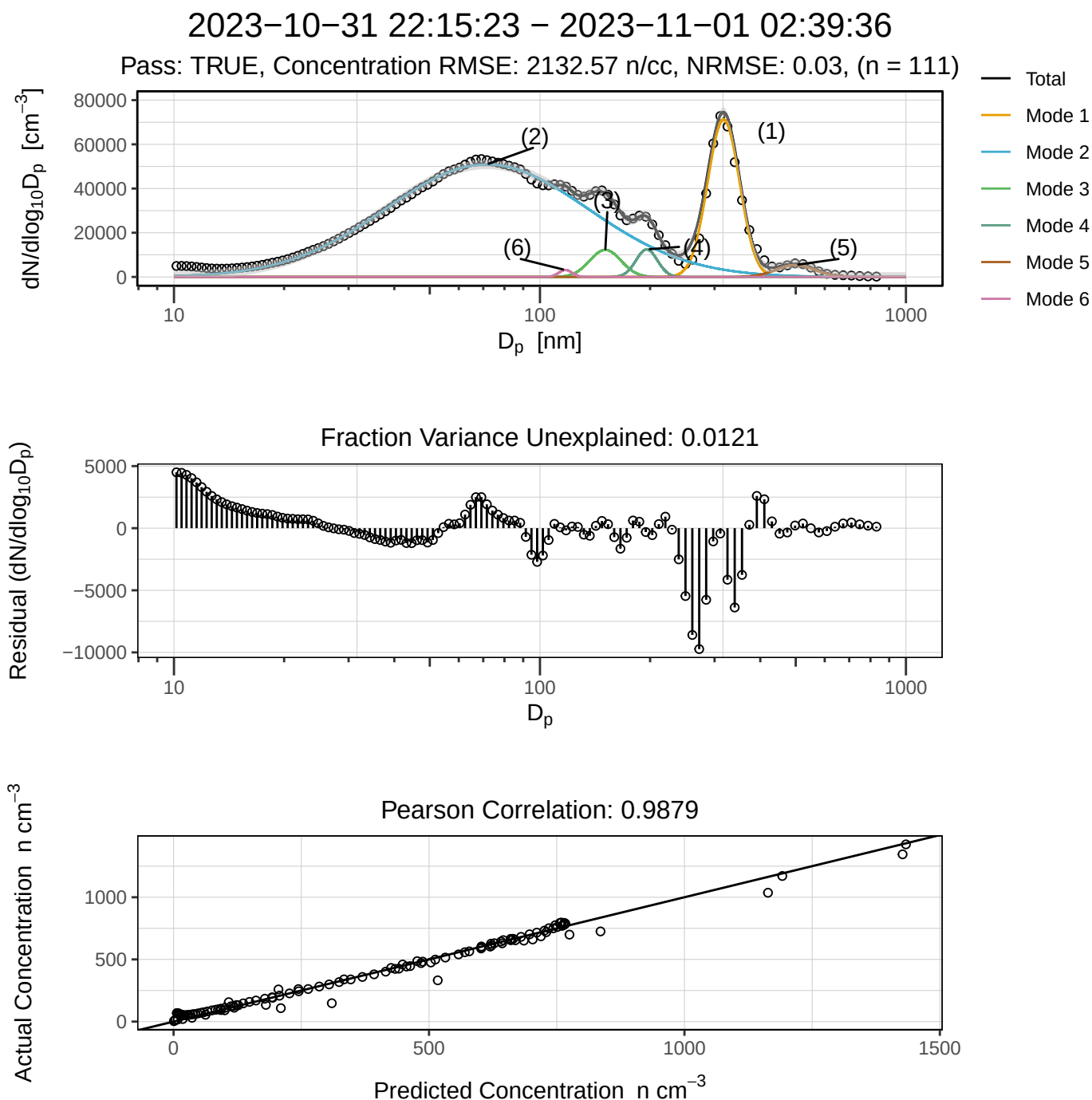


Figure S1 - An example of how to use the multimodal algorithm on example laboratory data. Note this reproduces the full version of Figure 2 in the manuscript.

The remaining outputs are data.frames of the combined actual/predicted data (Table S1), the data used to plot curves (Table S2), lognormal parameters of the modes (Table S3), and text output of the evaluation parameters.

Table S1 - Actual and predicted data for each diameter. Here we only present the first 6 rows of data for brevity. Note units and proper notation are missing due to restrictions for data.frame objects. From left to right: particle diameter (nm), predicted concentration ($dN/d\log D_p$), predicted number concentration (n cm⁻³), actual measurement concentration ($dN/d\log D_p$), actual number concentration (n cm⁻³), residual

particle concentration (dN/dlogDp), residual number concentration (n cm⁻³), and ratio of actual to predicted concentration for each bin.

Dp	Predicted dNdlogDp	Predicted dN	Actual dNdlogDp	Actual dN	Residual dNdlogDp	Residual dN	Ratio
10.16	453.91	6.30	4952.51	68.75	4498.60	62.45	0.09
10.49	529.09	7.12	4985.42	67.06	4456.33	59.95	0.11
10.82	612.31	8.47	4903.32	67.79	4291.01	59.33	0.12
11.17	709.75	9.78	4740.17	65.30	4030.42	55.52	0.15
11.53	820.21	11.25	4507.12	61.83	3686.91	50.58	0.18
11.90	944.94	12.90	4250.88	58.03	3305.94	45.13	0.22

Table S2 - Predicted concentration (dN/dlogD_p) for each identified mode to plot fitted modes. From left to right: particle diameter (nm), modes 1-6 concentration (dN/dlogD_p), and total concentration in (dN/dlogD_p).

Dp	Mode 1	Mode 2	Mode 3	Mode 4	Mode 5	Mode 6	dNdlogDp
10.00	0	420.26	0	0	0	0	420.26
10.01	0	422.31	0	0	0	0	422.31
10.02	0	424.37	0	0	0	0	424.37
10.03	0	426.43	0	0	0	0	426.43
10.04	0	428.50	0	0	0	0	428.50
10.05	0	430.58	0	0	0	0	430.58

Table S3 - Lognormal parameter output. From left to right: mode label, total number concentration (n cm⁻³), geometric standard deviation, geometric mean diameter (nm), maximum measured concentration for identified peak (dN/dlogD_p), lower peak diameter limit (nm), diameter of peak (nm), upper peak diameter limit (nm), width of identified peak in bins, Bayesian Information Criterion of model, . Note rounding is not applied by default.

Mode Label	N	GSD	Dpg	Max	Lower	Mode	Upper	Width
Mode 1	7710	1.10	318.0	72800	250.0	311.0	452	13
Mode 2	35200	1.89	71.6	53300	13.5	69.2	106	62
Mode 3	1300	1.10	151.0	12500	127.0	148.0	173	8
Mode 4	907	1.07	196.0	12100	166.0	195.0	272	12
Mode 5	715	1.13	499.0	5630	430.0	498.0	831	13
Mode 6	141	1.04	117.0	2930	98.3	118.0	137	9

BIC	RSS	TSS	R2	N	T	pval	GSD	T	pval	Dpg	T	pval
NA	1.50e+08	8.17e+09	0.982		0e+00			0				0
NA	8.86e+07	2.00e+10	0.996		0e+00			0				0
NA	4.78e+06	9.37e+07	0.949		8e-07			0				0
NA	2.10e+08	6.18e+08	0.660		4e-03			0				0
NA	1.49e+06	5.30e+07	0.972		0e+00			0				0
NA	1.39e+07	3.29e+07	0.576		4e-02			0				0

Table S4 - Statistical evaluation of the summed mode fits compared to the actual measurement. Pearson correlation coefficient, root mean square error of concentration (dN/dlogD_p) , max-min normalized root mean square error concentration (dN/dlogD_p), root mean square error of number concentration (n cm⁻³) ,

max-min normalized root mean square error of number concentration (n cm^{-3}), output of Student's t-test, and Chi-squared goodness-of-fit. Note as mentioned in the article, significance or goodness-of-fit testing did not demonstrate effectiveness in evaluating the model performance.

```
knitr::kable(round(result[[6]], 2))
```

Pearson Correlation	RMSE	NRMSE	dN	dN	Students T Test	Chi-Squared
			RMSE	NRMSE		
0.99	2132.57	0.03	37.15	0.03	0.85	0.24

3.2 Example 2 - Storm Peak Laboratory

```
## [1] "Log Path: /var/folders/2j/rmf9p1l50wg904hz8qkpv4m0000gn/T//Rtmpv6RxSf/multimodal20220323020341"
## [1] "2022-03-23 06:03:41: Error, please modify lower and upper limits to accommodate data set"
```

Notice the failure message? This is because the dataset begins for bin diameter 9.14. Now we can retry with adjusted limits.

```
## [1] "Log Path: /var/folders/2j/rmf9p1l50wg904hz8qkpv4m0000gn/T//Rtmpv6RxSf/multimodal20220323020341"
```

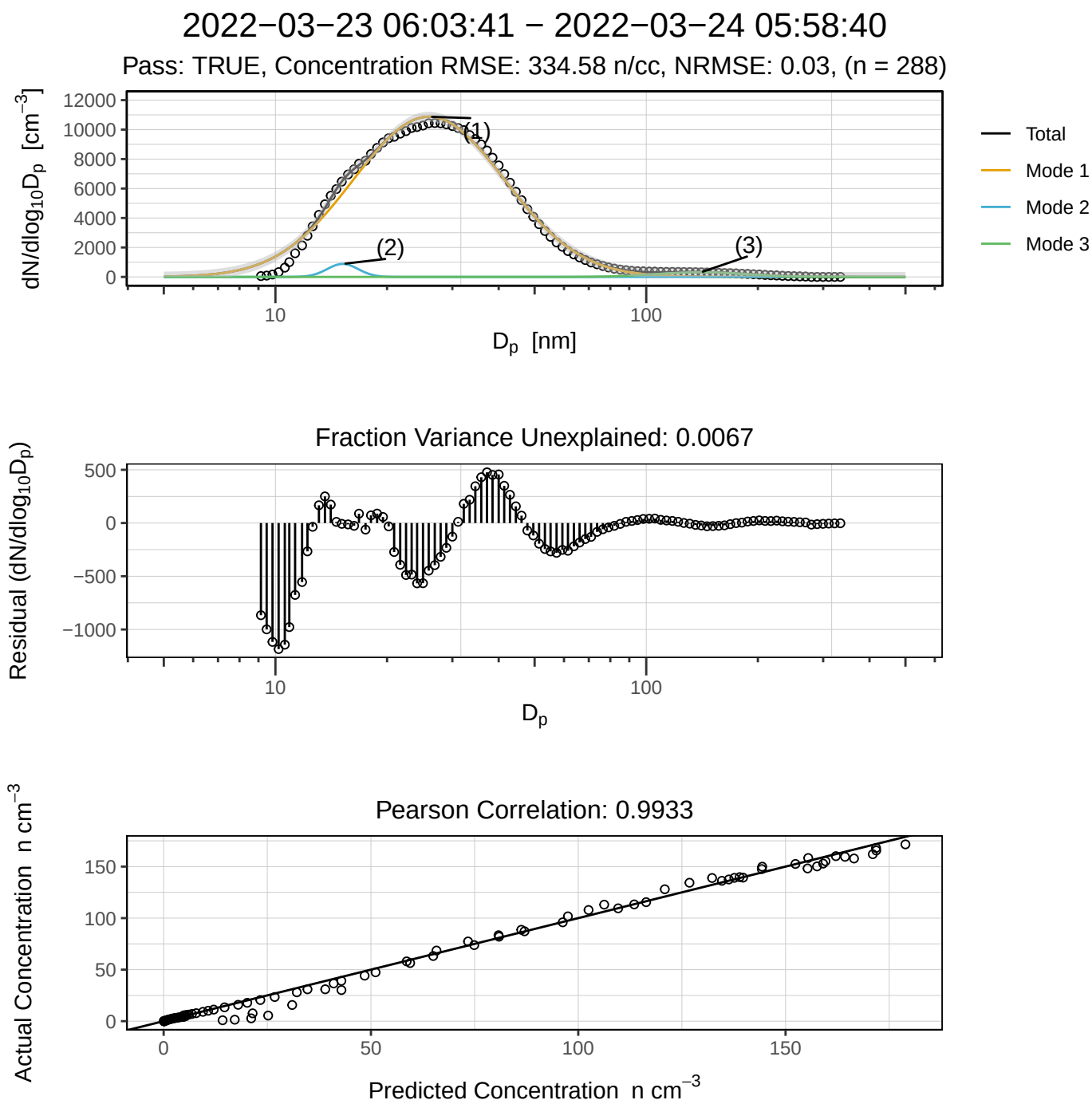


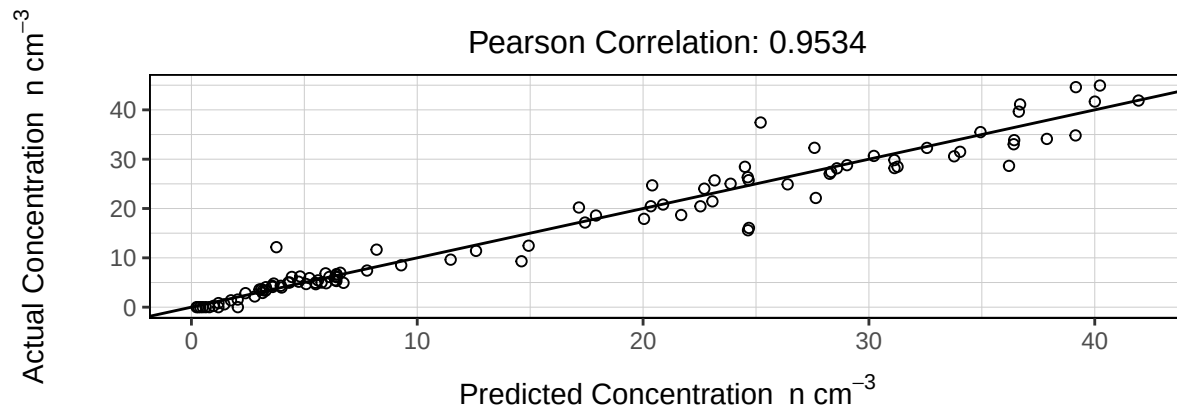
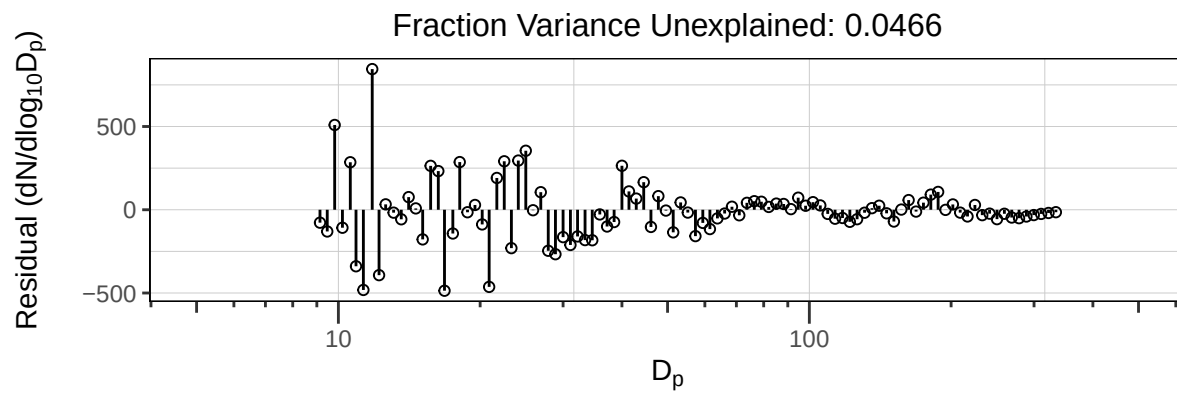
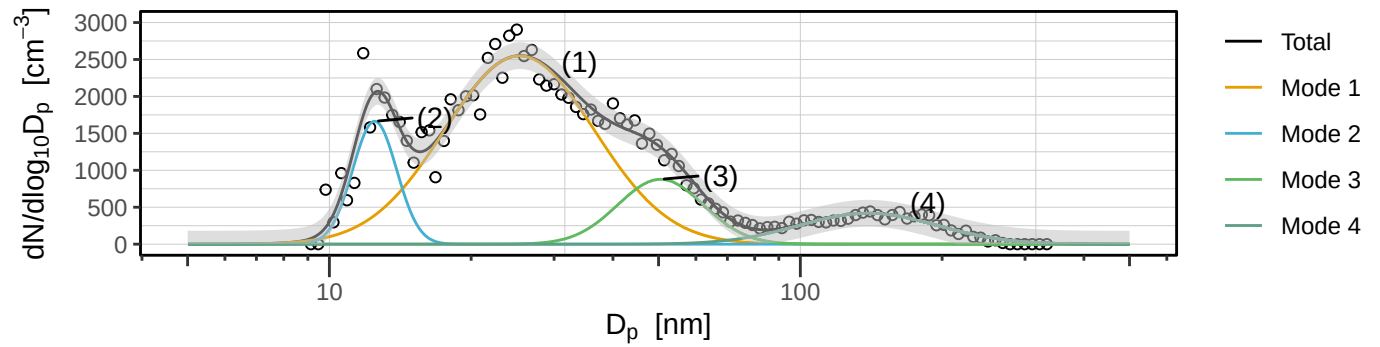
Figure 2 - Example of applying the multimodal.fitting function to a day of SPL averaged from 6AM to 6 PM. Note with the high averaging time a small peak appears where NPF events skew the average distribution between 10 - 20 nm.

Within this example TSI file for SPL there is a NPF event, but currently the averaging across the entire day removes all temporal variation. We will instead select times between 10:00 and 11:00 local time where the NPF event is just beginning and use high frequency data with smoothing to account for noise. We will only show two examples.

```
## [1] "Log Path: ~/Library/CloudStorage/Box-Box/Multimodal Curve Fitting/log//multimodal20220323110341."
```

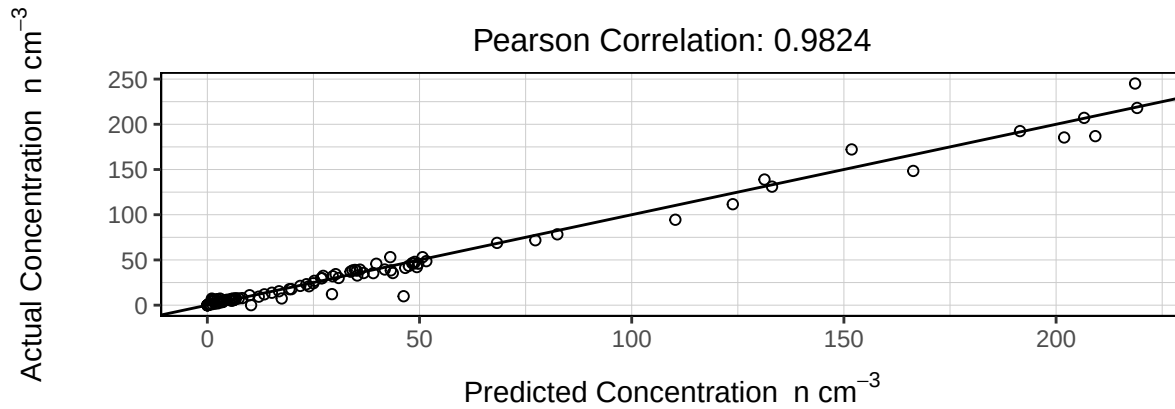
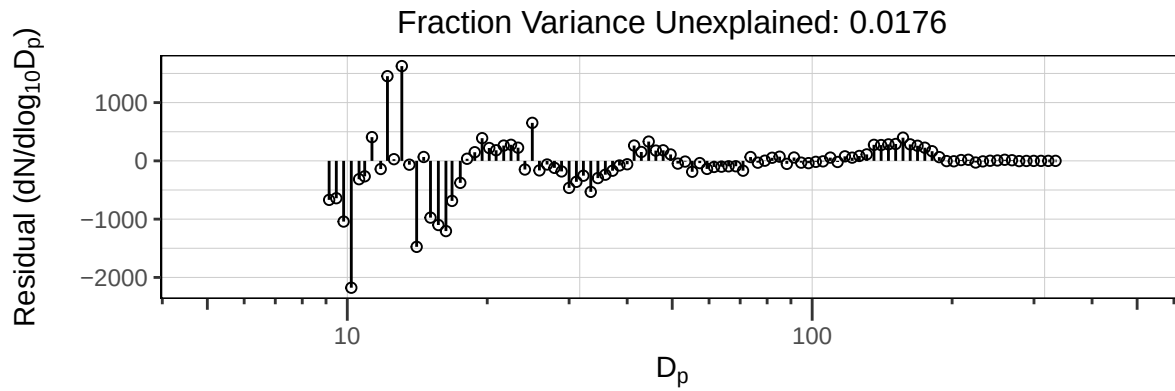
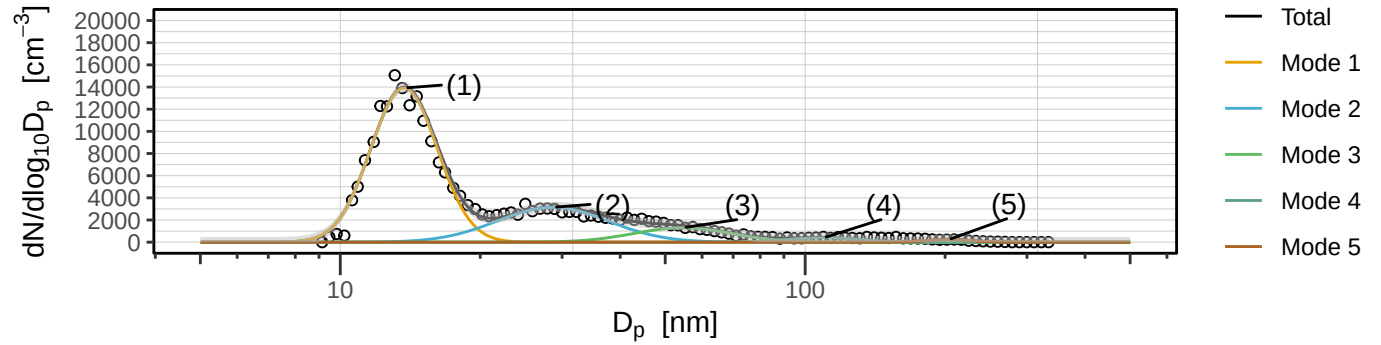

2022-03-23 15:18:41 – 2022-03-23 15:18:41

Pass: TRUE, Concentration RMSE: 181.09 n/cc, NRMSE: 0.06, (n = 1)



2022-03-23 15:53:41 – 2022-03-23 15:53:41

Pass: TRUE, Concentration RMSE: 458.35 n/cc, NRMSE: 0.03, (n = 1)



Now for the hour of data, 9 of the 12 fittings for the beginning of the NPF event (75% were successful) (i.e. explained 90% of concentration variance and had a NRMSE of 0.1).